

Safety assessment of the process KREYENBORG IR Clean+ (universal) used to recycle post-consumer PET into food contact materials

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The declarations of interest of all scientific experts active in EFSA's work are available at <https://open.efsa.europa.eu/experts>

Abstract

The EFSA Panel on Food Contact Materials (FCM) assessed the safety of the recycling process KREYENBORG IR Clean+ (universal) (EU register number RECYC329). The input is hot washed and dried poly(ethylene terephthalate) (PET) flakes mainly originating from collected post-consumer PET containers, with no more than 5% PET from non-food consumer applications. The flakes are heated in a continuous infrared (IR) dryer (step 2) before being processed in a finisher reactor (step 3). Having examined the challenge test provided, the Panel concluded that both steps are critical in determining the decontamination efficiency of the process. The operating parameters to control the efficiency of these critical steps are temperature, air/PET ratio and residence time. It was demonstrated that this recycling process ensures that the level of migration of potential unknown contaminants into food is below the conservatively modelled migration of 0.0481 or 0.0962 µg/kg food, depending on the molar mass of a contaminant substance. Therefore, the Panel concluded that the recycled PET obtained from this process is not of safety concern, when used at up to 100% for the manufacture of materials and articles for contact with all types of foodstuffs, including drinking water, for long-term storage at room temperature or below, with or without hot-fill. Articles made of this recycled PET are not intended to be used in microwave and conventional ovens and such uses are not covered by this evaluation.

KEYWORDS

food contact materials, Kreyenborg GmbH & Co. KG., plastic, poly(ethylene terephthalate) (PET), recycling process, safety assessment

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1 | INTRODUCTION

1.1 | Background

Recycled plastic materials and articles shall only be placed on the market if the recycled plastic is from an authorised recycling process. Before a recycling process is authorised, the European Food Safety Authority (EFSA)'s opinion on its safety is required. This procedure has been established in Articles 17 and 18 of Commission Regulation (EU) 2022/1616¹ on recycled plastic materials intended to come into contact with foods. More specifically, according to Article 18 of Commission Regulation (EU) 2022/1616 on recycled plastic materials intended to come into contact with foods, EFSA is required to carry out risk assessments on the risks originating from the migration of substances from recycled food contact plastic materials and articles into food, to evaluate the microbiological safety of these materials and articles and to deliver a scientific opinion on the recycling process examined.

According to this procedure, the process developers submit applications to the competent authorities of Member States, which transmit the applications to EFSA for evaluation. In this case, EFSA received an application from the German Competent Authority (Bundesamt für Verbraucherschutz und Lebensmittelsicherheit), for evaluating the recycling process KREYENBORG IR Clean+ (universal European Union (EU) register No RECYC329). The request has been registered in the EFSA's register of received questions under the number EFSA-Q-2023-00464. The dossier was submitted by Kreyenborg GmbH & Co. KG, Messingweg 18, 48308 Senden, Germany (see ['Documentation provided to EFSA'](#)).

1.2 | Terms of Reference

The German Competent Authority (Bundesamt für Verbraucherschutz und Lebensmittelsicherheit) requested the safety evaluation of the recycling process KREYENBORG IR Clean+ (universal), in compliance with Article 17 of Commission Regulation (EU) 2022/1616. The recycling process uses the recycling technology number 1 of the list of suitable recycling technologies of Table 1 of Annex 1 of Commission Regulation (EU) 2022/1616.

2 | DATA AND METHODOLOGIES

2.1 | Data

The applicant submitted a confidential and a non-confidential version of a dossier, following EFSA's 'Scientific Guidance on the criteria for the evaluation and on the preparation of applications for the safety assessment of post-consumer mechanical PET recycling processes intended to be used for manufacture of materials and articles in contact with food' (EFSA CEP Panel, 2024) and EFSA's 'Administrative guidance for the preparation of applications for the authorisation of individual recycling processes to produce recycled plastics materials and articles intended to come into contact with food' (EFSA, 2024).

Additional information was received from the applicant during the assessment process, in response to a request from EFSA sent on 25 March 2024. Following a clarification teleconference held on 17 January 2025, additional data were spontaneously submitted by the applicant on 4 February 2025 (see ['Documentation provided to EFSA'](#)).

In accordance with Art. 38 of the Regulation (EC) No 178/2002² and taking into account the protection of confidential information and the personal data in accordance with Articles 39 to 39e of the same Regulation, and of the Decision of the EFSA's Executive Director laying down practical arrangements concerning transparency and confidentiality,³ the non-confidential version of the dossier has been published on Open.EFSA.⁴

According to Art. 32c(2) of Regulation (EC) No 178/2002 and to the Decision of EFSA's Executive Director laying down the practical arrangements on pre-submission phase and public consultations, EFSA carried out a public consultation on the non-confidential version of the application from 24 September to 15 October 2024 for which no comments were received.

The following information on the recycling process was provided by the applicant and used for the evaluation (EFSA, 2024; EFSA CEP Panel, 2024):

- Recycling process,
- Determination of the decontamination efficiency of the recycling process,
- Table of operating parameters,
- Self-evaluation of the recycling process.

¹Commission Regulation (EU) 2022/1616 of 15 September 2022 on recycled plastic materials and articles intended to come into contact with foods, and repealing Commission Regulation (EC) No 282/2008. OJ L 243, 20.9.2022, p. 3–46.

²Commission Regulation (EC) No 178/2002 of the European Parliament and of the Council of 28 January 2002 laying down the general principles and requirements of food law, establishing the European Food Safety Authority and laying down procedures in matters of food safety. OJ L 31, 1.2.2002, p. 1–48.

³Decision available at: <https://www.efsa.europa.eu/en/corporate-pubs/transparency-regulation-practical-arrangements>.

⁴The non-confidential version of the dossier has been published on Open.EFSA and is available at the following link: <https://open.efsa.europa.eu/dossier/FCM-2023-17294>.

2.2 | Methodologies

The risks associated with the use of recycled plastic materials and articles in contact with food come from the possible migration of chemicals into the food in amounts that would endanger human health. The quality of the input, the efficiency of the recycling process to remove contaminants as well as the intended use of the recycled plastic are crucial points for the risk assessment (EFSA CEP Panel, 2024).

The criteria for the safety evaluation of a mechanical recycling process to produce recycled PET intended to be used for the manufacture of materials and articles in contact with food are described in the scientific guidance developed by the EFSA Panel on Food Contact Materials, Enzymes and Processing Aids (EFSA CEP Panel, 2024). The principle of the evaluation is to apply the decontamination efficiency of a recycling process, obtained from a challenge test with surrogate contaminants, to a reference contamination level for post-consumer PET, conservatively set at 3 mg/kg PET for contaminants resulting from possible misuse. The resulting residual concentration of each surrogate contaminant in recycled PET (C_{res}) is compared with a modelled concentration of the surrogate contaminants in PET (C_{mod}). This C_{mod} is calculated using generally recognised conservative migration models so that the related migration does not give rise to a dietary exposure exceeding 0.0025 µg/kg body weight (bw) per day (i.e. the human exposure threshold value for chemicals with structural alerts for genotoxicity), below which the risk to human health would be negligible, considering different dietary exposure scenarios (EFSA CEP Panel, 2024). If the C_{res} is not higher than the C_{mod} , the recycled PET manufactured by such recycling process is not considered of safety concern for the defined conditions of use (EFSA CEP Panel, 2024).

The assessment was conducted in line with the principles described in the EFSA Guidance on transparency in the scientific aspects of risk assessment, considering the relevant guidance from the EFSA Scientific Committee (EFSA, 2009).

3 | ASSESSMENT

3.1 | General information⁵

According to the applicant, the recycling process KREYENBORG IR Clean+ (universal) is intended to recycle food grade PET containers. The recycled PET is intended to be used at up to 100% for the manufacture of materials and articles for direct contact with all kinds of foodstuffs, such as thermoformed trays/containers, take-away packaging and bottles for mineral water, soft drinks, juices and beer, for long-term storage at room temperature or below, with or without hot-fill. The final articles are not intended to be used in microwave or conventional ovens.

3.2 | Description of the process

3.2.1 | General description⁶

The recycling process KREYENBORG IR Clean+ (universal) produces recycled PET flakes from PET materials originating from post-consumer collection systems (kerbside and deposit collection systems).

Input

- In step 1, the post-consumer PET is sorted and processed into hot washed and dried flakes.

The decontamination process comprises the two steps below.

Decontamination and production of recycled PET material

- In step 2, the flakes are [REDACTED] by means of an infrared (IR) [REDACTED] dryer and decontaminated by air flow.
- In step 3, the flakes are further decontaminated in a finisher reactor under [REDACTED] air flow and [REDACTED].

The operating conditions of the process have been provided to EFSA.

⁵Technical dossier, section 'Intended application in contact with food'.

⁶Technical dossier, sections 'Recycling process', 'Characterisation of the input' and 'Characterisation of the recycled plastic'.

3.2.2 | Characterisation of the pre-processed plastic input⁷

According to the applicant, the input material consists of hot washed and dried flakes obtained from PET materials, e.g. bottles, previously used for food packaging, from post-consumer collection systems (kerbside and deposit systems). A small fraction may originate from non-food applications. According to the applicant, the proportion will be no more than 5%, as specified in Article 7 and Table 1 of Annex I of Commission Regulation (EU) 2022/1616.

Technical specifications on the hot washed and dried flakes are provided, such as on physical properties and residual contents of moisture, poly(vinyl chloride) (PVC), polyamide (PA), polyolefins, glue, cellulose and metals (see Appendix A).

3.3 | KREYENBORG IR Clean+ (universal) process

3.3.1 | Description of the main steps⁸

The process flow diagram of the KREYENBORG IR Clean+ (universal) process, as provided by the applicant, is reported in Figure 1. The steps are:

- Decontamination by means of an IR dryer (step 2):

The IR dryer consists of a horizontal rotating drum with an IR heater module in the centre of the drum. The flakes are continuously introduced and treated by IR radiation under defined conditions of air flow, temperature profile and residence time.

- Decontamination of the flakes in a finisher reactor (step 3):

The finisher consists of a vertical drying hopper with hot air supply.

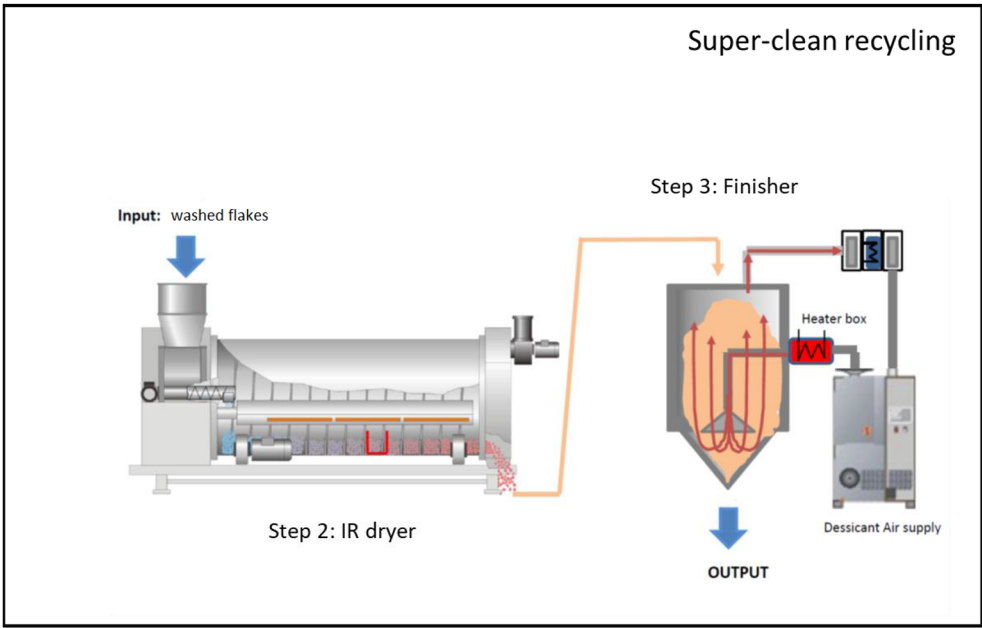


FIGURE 1 Process flow diagram provided by the applicant.

The process is run under defined operating parameters⁹ of temperature, air/PET ratio and residence time. All critical parameters are monitored by sensors provided in the piping and instrumentation diagram of the application. The values are automatically reported via the relevant software, alerting in case of deviations from the acceptable parameter values or ranges.¹⁰

Flakes, the final product of the process, are checked against technical requirements, such as intrinsic viscosity, colour and black spots.

⁷Technical dossier, section 'Characterisation of the input'.
⁸Technical dossier, sections 'Recycling process' and 'Determination of the decontamination efficiency of the recycling process'.
⁹In accordance with Art. 9 and 20 of Regulation (EC) No 1935/2004, the parameters were provided to EFSA by the applicant and made available to the Member States and the European Commission (see Appendix C).
¹⁰Technical dossier, Section 'Quality Assurance System'.

3.3.2 | Decontamination efficiency of the recycling process¹¹

To demonstrate the decontamination efficiency of the recycling process, a challenge test performed on steps 2 and 3 was submitted to EFSA.

PET flakes were contaminated with toluene, chlorobenzene, chloroform, methyl salicylate, phenylcyclohexane, benzophenone and methyl stearate, selected as surrogates in agreement with the EFSA Scientific Guidance (EFSA CEP Panel, 2024) and in accordance with the recommendations of the US Food and Drug Administration (FDA, 2021).

A mixture of solid surrogates (benzophenone and methyl stearate) and liquid surrogates (toluene, chlorobenzene, chloroform, methyl salicylate and phenylcyclohexane) was added in a barrel to 25 kg of conventionally recycled¹² post-consumer PET flakes. Four such barrels were prepared and stored for 7 days at 50°C with periodical agitation. Afterwards, the contaminated flakes were rinsed with 10% ethanol and air-dried. For each barrel, the concentration of surrogates was determined before and after air-drying. The barrels were merged into a single batch of 100 kg.

Steps 2 and 3 were challenged in an industrial-scale installation. According to the applicant, samples were analysed after each step. Sampling data submitted met the requirements of the EFSA Scientific Guidance (EFSA CEP Panel, 2024) Art. 5.2.3:

- contaminated flakes
- washed and dried flakes
- after the IR dryer
- after the finisher.

The decontamination efficiency was calculated from the concentration differences of the surrogate substances in the flakes sampled before the IR dryer (step 2) and after the finisher (step 3). The results are summarised in Table 1.

TABLE 1 Efficiency of the decontamination of the KREYENBORG IR Clean+ (universal) process in the challenge test.

Surrogates	Concentration of surrogates before step 2 (mg/kg PET)	Concentration of surrogates after step 3 (mg/kg PET)	Decontamination efficiency (%)
Toluene	162.7	< 0.1 ^a	> 99.9
Chlorobenzene	330.8	1.4	99.6
Chloroform	113.7	1.1	99.0
Methyl salicylate	411.4	4.3	99.0
Phenylcyclohexane	294.5	6.6	97.8
Benzophenone	617.1	30.7	95.0
Methyl stearate	798.3	31.7	96.0

Abbreviation: PET, poly(ethylene terephthalate).
^aNot detected at the limits of detection given.

3.4 | Discussion

Considering the high temperatures used during the process, the possibility of contamination by microorganisms can be discounted. Therefore, this evaluation focuses on the chemical safety of the final product.

Specifications on the input material (i.e. washed and dried flakes, step 1) are listed in Appendix A.

The flakes are produced from PET containers, e.g. bottles, previously used for food packaging, collected through post-consumer collection systems. However, a small fraction may originate from non-food applications, such as bottles for soap, mouthwash or kitchen hygiene agents. According to the applicant, the collection system and the sorting are managed in such a way that this fraction will be no more than 5% in the input stream, as recommended by the EFSA CEP Panel in its Guidance (EFSA CEP Panel, 2024).

The KREYENBORG IR Clean+ (universal) process is adequately described. It comprises the IR dryer (step 2) and the finisher (step 3). The operating parameters of temperature, residence time and air/PET ratio have been provided to EFSA.

A challenge test to measure the decontamination efficiency was conducted on process steps 2 and 3 using a small industrial-scale installation. The Panel considered that it was performed correctly according to the recommendations of the EFSA Guidance (EFSA CEP Panel, 2024). The challenge test was considered representative of the larger industrial installations, provided the throughput increases in relation to the geometric characteristics. The operating conditions applied in the challenge test were equivalent or less severe than in the commercial process of Appendix C. Consequently, the temperature, residence time and air/PET ratio specified in Appendix C are to be controlled to guarantee the efficiency of the decontamination.

¹¹Technical dossier, Section ‘Determination of the decontamination efficiency of the recycling process’.
¹²Conventional recycling commonly includes sorting, grinding, washing and drying steps. It produces washed and dried flakes.

The decontamination efficiencies obtained for each surrogate, ranging from 95.0% to > 99.9%, have been used to calculate the residual concentrations of potential unknown contaminants in PET (C_{res}). By applying the decontamination efficiency percentage to the reference contamination level of 3 mg/kg PET, the C_{res} values shown in Table 2 were obtained.

According to the evaluation principles (EFSA CEP Panel, 2024), the dietary exposure must not exceed 0.0025 $\mu\text{g/kg}$ bw per day, below which the risk to human health is considered negligible. The C_{res} value should not exceed the modelled concentration in PET (C_{mod}) that, after 1 year at 25°C, results in a migration giving rise to a dietary exposure of 0.0025 $\mu\text{g/kg}$ bw per day. As the recycled PET is intended for the manufacturing of articles (e.g. bottles) to be used in direct contact with drinking water, the exposure scenario for infants has been applied for the calculation of C_{mod} (Exposure Scenario A; water could be used to prepare infant formula). A maximum dietary exposure of 0.0025 $\mu\text{g/kg}$ bw per day corresponds to a maximum migration of 0.0481 $\mu\text{g/kg}$ ($= 5 \times 0.00962 \mu\text{g/kg}$) or 0.0962 $\mu\text{g/kg}$ ($= 10 \times 0.00962 \mu\text{g/kg}$), depending on the molar mass of a contaminant substance into infant's food and has been used to calculate C_{mod} (EFSA CEP Panel, 2024). C_{res} reported in Table 2 is calculated for 100% recycled PET. The results of these calculations are shown in Table 2. The relationship between the key parameters for the evaluation scheme is reported in Appendix B.

TABLE 2 Decontamination efficiency from the challenge test, residual concentrations of the surrogates (C_{res}) related to the reference contamination level and calculated concentrations of the surrogates in PET (C_{mod}) corresponding to a modelled migration of 0.0481 or 0.0962 $\mu\text{g/kg}$ after 1 year at 25°C (C_{mod}).

Surrogates	Decontamination efficiency (%)	C_{res} for 100% rPET (mg/kg PET)	C_{mod} (mg/kg PET) Scenario A
Toluene	> 99.9	< 0.003	0.04
Chlorobenzene	99.6	0.01	0.05
Chloroform	99.0	0.03	0.05
Methyl salicylate	99.0	0.03	0.12
Phenylcyclohexane	97.8	0.07	0.13
Benzophenone	95.0	0.15	0.15
Methyl stearate	96.0	0.12	0.29

Abbreviations: PET, poly(ethylene terephthalate); rPET, recycled poly(ethylene terephthalate).

On the basis of the provided data from the challenge test and the applied conservative assumptions, the Panel considered that under the given operating conditions the recycling process KREYENBORG IR Clean+ (universal) is able to ensure that the level of migration of unknown contaminants from the recycled PET into infant's food is below the conservatively modelled migration of 0.0481 or 0.0962 $\mu\text{g/kg}$, depending on the molar mass of a contaminant substance. At this level, the risk to human health is considered negligible when the recycled PET is used at up to 100% to produce materials and articles intended for contact with all types of foodstuffs, including drinking water (exposure Scenario A), for long-term storage at room temperature or below, with or without hot-fill.

4 | CONCLUSIONS

The Panel considered that the process KREYENBORG IR Clean+ (universal) is adequately characterised and that the main steps used to recycle the PET flakes into decontaminated PET pellets have been identified. Having examined the challenge test provided, the Panel concluded that the IR dryer (step 2) and the finisher reactor (step 3) are critical for the decontamination efficiency. The parameters to control the process performance are temperature, residence time and air/PET ratio as specified in Appendix C.

The Panel concluded that the recycling process KREYENBORG IR Clean+ (universal) is capable of reducing contamination of post-consumer food contact PET to a concentration that does not give rise to concern for a risk to human health if:

- (i) it is operated under conditions that are at least as severe as those applied in the challenge test used to measure the decontamination efficiency of the process;
- (ii) the input material of the process is washed and dried post-consumer PET flakes originating from materials and articles that have been manufactured in accordance with the EU legislation on food contact materials and contain no more than 5% of PET from non-food consumer applications;
- (iii) the recycled PET is used at up to 100% for the manufacture of materials and articles for contact with all types of foodstuffs, including drinking water, for long-term storage at room temperature or below, with or without hot-fill.

The final articles made of this recycled PET are not intended to be used in microwave and conventional ovens and such uses are not covered by this evaluation.

5 | RECOMMENDATION

The Panel recommended periodic verification that the input to be recycled originates from materials and articles that have been manufactured in accordance with the EU legislation on food contact materials and that the proportion of PET from non-food consumer applications is no more than 5%. This adheres to good manufacturing practice and the Commission Regulation (EU) 2022/1616. Critical steps in recycling should be monitored and kept under control. In addition, supporting documentation should be available on how it is ensured that the critical steps are operated under conditions at least as severe as those in the challenge test used to measure the decontamination efficiency of the process.

6 | DOCUMENTATION PROVIDED TO EFSA

Dossier 'KREYENBORG IR Clean+ (universal)'. December 2023. Submitted by Kreyenborg GmbH & Co. KG., Germany.

Additional information, October 2024. Submitted by Kreyenborg GmbH & Co. KG., Germany.

Additional data, February 2025. Submitted by Kreyenborg GmbH & Co. KG., Germany.

ABBREVIATIONS

bw	body weight
CEP	Panel on Food Contact Materials, Enzymes and Processing Aids
C_{mod}	modelled concentration in PET
C_{res}	residual concentration in PET
FCM	Panel on Food Contact Materials
IR	infrared
PET	poly(ethylene terephthalate)
rPET	recycled poly(ethylene terephthalate)

REQUESTOR

German Competent Authority, Bundesamt für Verbraucherschutz und Lebensmittelsicherheit, Germany.

QUESTION NUMBER

EFSA-Q-2023-00464

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PANEL MEMBERS

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WAIVER

In accordance with Article 21 of the Decision of the Executive Director on Competing Interest Management a waiver was granted to an expert of the Working Group. Pursuant to Article 21(6) of the aforementioned Decision, the concerned expert was allowed to take part in the preparation and discussion of the scientific output but was not allowed to take up the role of rapporteur within that time frame. Any competing interests are recorded in the respective minutes of the meetings of the FCM Panel Working Group on Recycling Plastics.

LEGAL NOTICE

Relevant information or parts of this scientific output have been blackened in accordance with the confidentiality requests formulated by the applicant pending a decision thereon by EFSA. The full output has been shared with the European Commission, EU Member States (if applicable) and the applicant. The blackening may be subject to review once the decision on the confidentiality requests is adopted by EFSA and in case it rejects some of the confidentiality requests.

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FDA (Food and Drug Administration). (2021). Guidance for industry: Use of recycled plastics in food packaging: Chemistry considerations. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/guidance-industry-use-recycled-plastics-food-packaging-chemistry-considerations>.

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APPENDIX A

Specifications of the pre-processed input material as provided by the applicant¹³

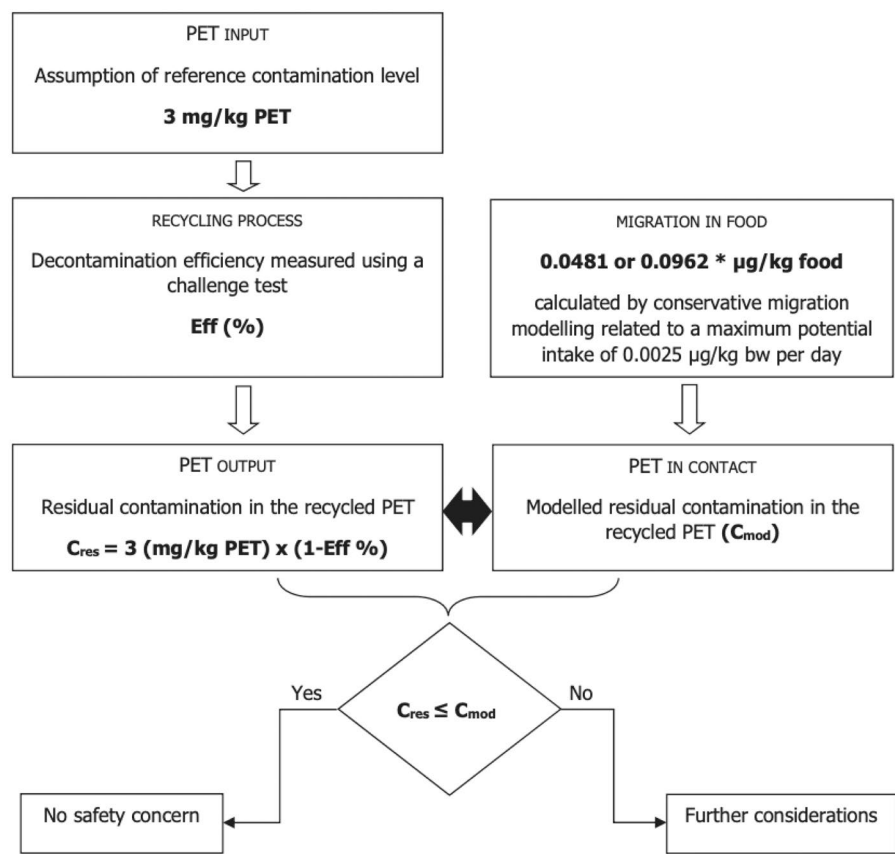
Parameter	Value
Moisture max.	1.0%
Moisture variation	± 0.3%
Bulk density	350–400 kg/m ³
Bulk density variation	± 150 kg/m ³ /h
Material temperature	5–40°C
Material temperature variation	± 10°C/h
PVC max.	100 mg/kg
Glue max.	100 mg/kg
Polyolefins max.	100 mg/kg
Cellulose (paper, wood)	100 mg/kg
Metals max.	50 mg/kg
Polyamide max.	50 mg/kg

Abbreviation: PVC, poly(vinyl chloride).

¹³Technical dossier, section ‘Characterisation of the input’.

APPENDIX B

Relationship between the key parameters for the evaluation scheme, based on the most conservative scenario A (EFSA CEP Panel, 2024)



*The figures are derived from the application of the human exposure threshold value of 0.0025 µg/kg bw per day applying the factors of 5 and 10 related to the overestimation of modelling (most conservative Scenario A).

*Depending on the molecular mass of the surrogate substance.

APPENDIX C

Table of operational parameters¹⁴

Although the operational parameters are reported for all the process steps, the critical steps and the corresponding parameters of the challenge test/process, considered for the evaluation of the process and for which it has been concluded that the process is safe, are highlighted in green.

The process should be operated at conditions at least as severe as the ones indicated in green in the table (e.g. lower pressures are more severe than higher, higher temperatures are more severe than lower, longer times are more severe than shorter, higher gas flows generally are more severe than lower).

The official enforcement control shall verify that the recycling plant is operating in a way that complies with its authorisation. Depending on the technology, some of the parameters are inter-related and changing one parameter to a more severe value may impact another parameter into a less severe value. Therefore, eventual deviations from the values of the parameters indicated as critical (marked in green in the table) should be demonstrated not impacting significantly on the safety assessment. The table does not necessarily report all the tolerances for the operational parameters.

Recycling process KREYENBORG IR Clean+ (universal) (RECYC329)						
Parameters	Step 2: IR Dryer			Step 3: Finisher reactor		
	t (min)	T (°C)	Air/PET ratio (m ³ /h) per (kg/h)	t (min)	T (°C)	Air/PET ratio (m ³ /h) per (kg/h)
	continuous			continuous		
Process	≥ continuous	≥	≥	≥ continuous	≥	≥

Abbreviations: IR, infrared; PET, poly(ethylene terephthalate).

¹⁴Technical dossier, Sections ‘Table of operating Parameters’ and ‘Recycling process’.